

YEAR 7 · UNIT 2 · HOMEWORK 7

What Kind, and How Many

Mathematics 2.3–2.4 – collecting like terms, expanding brackets

Science 2.5–2.6 – atoms, elements, compounds and formulae

Section E – the two rules both subjects taught you this week

Name: _____

Class: _____

Date given: _____

Date due: **Monday 21 September 2026****How to do this homework**Five sections. Maths first, then science, then **Section E**, where the two meet.

- In maths, $3x + 5$ is a **finished answer**. You are allowed to stop.
- In science, write **full English sentences**. “Compound” is not a sentence; “Salt is a compound of sodium and chlorine” is.
- **No calculator.**

Section A · Mathematics 2.3 – Collecting Like Terms**A1 How many terms?**

Count the terms in each expression. The first one is done for you.

What a term is

A **term** is one part of an expression. The **plus and minus signs separate the terms**. A number on its own is a term too. So $4a + 3b$ has **two** terms: $4a$ and $3b$.

Expression	Terms	Expression	Terms
$4a + 3b$	2	$8m - 3m + n$	
$5x$		6	
$2p + 7q + 4$		$y + 2z - 5 + 3y$	

A2 Like or not like?

Tick one box in each row.

Word help – one word with two jobs

like every day, to enjoy something. In maths, **the same kind**: *milk is like water – both are liquids.*

like terms terms with **the same letter**, so they can be added together

	Like	Not like		Like	Not like
$3a$ and $5a$			$2ab$ and $5ba$		
$4x$ and $4y$			m and m^2		
7 and 2			$5c$ and $-2c$		

A3 Simplify

Collect the like terms. Two of these are already finished – leave them alone.

x means 1x

A letter on its own already has a **1** in front of it. Nobody writes the 1, but it is there.

$$8s - s = 8s - 1s = 7s \quad \text{not 8.}$$

(a) $2b + 3b =$ _____

(f) $4x + x + 2x =$ _____

(b) $9k - 4k =$ _____

(g) $3a + 7 =$ _____

(c) $6m + 3n + 2m =$ _____

(h) $6y - y + 2y =$ _____

(d) $5p + 2q + p + 4q =$ _____

(i) $10g - 6g - g =$ _____

(e) $8s - s =$ _____

(j) $5c + 3d =$ _____

A4 Every sign travels with its own term

Take the sign with you

The + or – in front of a term **belongs to that term**. When you move the term, the sign goes with it.

$$7x + 5y - 3x + y = 7x - 3x + 5y + y = 4x + 6y$$

Simplify. Write the middle line, with the like terms next to each other.

(a) $9x + 4y - 5x + 2y$ _____

(b) $8a - 3b + 2a - b$ _____

(c) $6 + 5t - 1 - 2t$ _____

(d) $7p - 2q - 3p + 6q + 4$

A5 Now you write the question**Read all four conditions before you start**

Here is an expression:

$5a + 3b$

Write a **word problem** that this expression is the answer to. Your question must:

1. be about something **real** — a shop, a classroom, a lab bench;
2. use **two different kinds** of thing, one for a and one for b ;
3. say clearly what a and b **stand for**;
4. be written in **full sentences** and **end in a question**.

Section B · Mathematics 2.4 — Expanding Brackets**B1 Fill the grid**

One box, one multiplication. Fill in the empty boxes, then write the answer.

Word help — the words that carry the whole rule

expand · multiply out the same instruction in different words. Both appear in the exercise

each / every not a maths word, but it carries the whole rule: **every** term inside, not just the first

Brackets mean multiply $4(x + 3)$ means $4 \times (x + 3)$. The times sign is hidden, exactly as it is in $4n$.

$\times$	x	$+ 3$
4	$4x$	$+ 12$

$4(x + 3) = 4x + 12$

Not $4x + 3$. The 3 is inside the brackets, so it gets multiplied too.

×	m	$+ 4$
3		

(a) $3(m + 4) =$ _____

×	y	$+ 2$
5		

(b) $5(y + 2) =$ _____

×	$2n$	$+ 3$
4		

(c) $4(2n + 3) =$ _____

B2 Expand

No grid this time. Draw one if it helps.

Two things that go inside with the term**The minus sign.** $3(x - 2) = 3x - 6$ and $7(1 - w) = 7 - 7w$.**A number already in front of the letter.** Multiply the numbers, keep the letter: $5 \times 2p = 10p$, **not** $52p$.

(a) $3(a + 2) =$ _____

(e) $8(7 + z) =$ _____

(i) $4(3 - t) =$ _____

(b) $5(b + 3) =$ _____

(f) $6(k - 3) =$ _____

(j) $3(2q + 5) =$ _____

(c) $9(y + 3) =$ _____

(g) $2(y - 4) =$ _____

(k) $4(5u - 1) =$ _____

(d) $4(2 + f) =$ _____

(h) $5(m - 6) =$ _____

(l) $6(1 + 2v) =$ _____

(m) This one has three terms inside. $8(6 + 4w - 3g) =$ _____

B3 Knowing when to stop

Tick one box in each row. If it can be collected, write the answer.

Expanding does not switch off Section A

After you expand, look for like terms. Often there are none.

 $4(3 - c) = 12 - 4c$, and 12 and $4c$ are **not like terms**. So $12 - 4c$ is **finished**.

Expression	Finished	Can be collected	If so, write it
$12 - 4c$			
$5x + 3x$			
$3a + 6$			
$8 + 2m + 5$			

B4 Mr Bowen's homework, again

One of these four is right. Find it, and correct the other three.

(a) $3(a + 4) = 3a + 4$ _____

(c) $2(5 - d) = 10 - 2d$ _____

(b) $4(2b - 5) = 8b - 9$ _____

(d) $6(m + 2) = 6m + 8$ _____

B5 Expand, then collect

Two jobs, in this order. Expand the brackets first, then look for like terms.

(a) $3(2y + 1) + 5y + 4$

(b) $2(a + 6) + 3(a + 1)$

B6 The fish tanks

Mr Bowen keeps **7** fish tanks in his kitchen. Every tank holds f fish and **2** rubber ducks.

(a) Write an expression with brackets for the number of things in the tanks altogether.

(b) Expand it.

(c) Rubber ducks are not alive. Write an expression for the number of **living** things in the tanks.

Section C · Science 2.5 — Atoms, Elements and the Periodic Table

C1 Use the Periodic Table in your book

Groups are numbered 1 to 8.

Element	Symbol	Period	Group	Metal or non-metal
magnesium				
chlorine				

C2 Match the word to its meaningWrite the correct **letter** in each box.

Word		Meaning
1. Atom		A A column in the Periodic Table.
2. Element		B A short way to write the name of an element.
3. The Periodic Table		C Elements such as iron, copper and aluminium.
4. Period		D A tiny piece of matter. Everything is made of them.
5. Group		E A very, very small tube made of carbon atoms.
6. Symbol		F A substance made of only one kind of atom.
7. Nanotube		G A row in the Periodic Table.
8. Metals		H A way of arranging all the elements.

C3 The capital-letter rule

(a) Four students wrote a symbol wrongly. Write it correctly, and name the element.

One capital, then small letters

An element symbol starts with a **CAPITAL** letter. If there is a second letter it is **small**.
Mg is magnesium — **one** element. MG and mg are not symbols at all.

Written	Correct	Element	Written	Correct	Element
NA			mG		
cl			CA		

(b) Mr Bowen means cobalt, and writes CO. Cobalt is Co. Explain what he has actually written down.

C4 Answer in full sentences

Diamond is hard and clear. Graphite is soft and grey. Both are made of **only** carbon atoms. Is diamond an element? Explain your answer.

Sentence starters — you may copy these

“An element is a substance that...” “...because every atom in it is...”

Section D · Science 2.6 – Compounds and Formulae

D1 Element or compound?

Tick one box in each row.

Count the kinds, not the atoms

An **element** is made of **one kind** of atom. A **compound** is made of **two or more kinds** of atom **bonded** together. O₂ has two atoms and they are **both oxygen** – one kind, so it is an **element**.

	Element	Compound		Element	Compound
Na			H ₂		
NaCl			CaCO ₃		
O ₂			Al		
CO ₂			H ₂ O		

D2 Naming a compound

Name the compound made from each set of elements.

Word help – the endings and the counting words

-ide	two elements joined. The metal comes first; the non-metal ends in -ide
-ate	two elements and oxygen joined. The ending is often -ate
mono / di	one / two. CO is carbon monoxide; CO ₂ is carbon dioxide

(a) sodium and chlorine

(c) lithium and sulfur

(b) magnesium and oxygen

(d) calcium, carbon and oxygen

(e) Mr Bowen writes *sulfur calcide*. Two things are wrong. Write the correct name and say what the two mistakes were.

D3 A compound is nothing like the elements in it

Sodium is a soft metal that bursts into flame in water. Chlorine is a poisonous yellow-green gas. Bonded together they make sodium chloride, which you ate with your dinner last night.

A student says: *"Salt must be dangerous, because chlorine is poisonous."* Explain, in full sentences, why this is wrong.

D4 Draw the particles

Draw the particles of each substance in the box. Use a different colour or pattern for each kind of atom, then tick element or compound.

oxygen O ₂	water H ₂ O	carbon dioxide CO ₂	methane CH ₄
element ☐ compound ☐	element ☐ compound ☐	element ☐ compound ☐	element ☐ compound ☐

Section E · Where the two subjects meet

In maths a letter stands for a number. In science a letter stands for an element. You read them with the same two rules.

The two rules

1. A letter says what kind. A number says how many. $3a \cdot O_3$
2. A number written in front multiplies everything after it. $4(x + 3) \cdot 3CO_2$

E1 Mg or mg?

Complete the table. The first row is done for you.

The capital is doing a job

In **maths** every letter is small, and two letters written together means **multiply**: mg means $m \times g$.

In **science** a symbol begins with a **capital**: Mg is magnesium, one element. Two capitals means **two** elements: CO is carbon and oxygen.

Written	Maths or science?	What it means
mg	maths	m multiplied by g
Mg		
CO		
Co		
xy		
$NaOH$		

E2 The 1 that nobody writes – in both subjects

Write the number that is **not printed**. The first one is done for you.

Same habit, two subjects

In **maths**, s means $1s$ – that is why $8s - s = 7s$. In **science**, H_2O has nothing after the O , so there is **one** oxygen atom. Both subjects leave the 1 out and expect you to know it is there.

Written	The question	The number
$8s - s$	how many s is the second term?	1
$w + w$	how many w is each term?	
H_2O	how many oxygen atoms?	
$NaCl$	how many chlorine atoms?	

E3 Counting atoms is expanding brackets

Complete the table. The first row is done for you.

The same grid, on a formula

$3CO_2$ means **three lots of** CO_2 . One CO_2 is one carbon atom and two oxygen atoms. So put it in the grid:

×	C	+ 2 O
3	3 C	+ 6 O

$$3(C + 2O) = 3C + 6O$$

Three CO₂ is **3 carbon atoms and 6 oxygen atoms**.

Formula	Atoms of each element	Atoms in total
3CO ₂	3 carbon, 6 oxygen	9
2H ₂ O		
4NaCl		
2CaCO ₃		

(b) Now write 2H₂O the way a maths teacher would. Fill the boxes.

$$2(2H + O) = \square H + \square O$$

E4 The same mistake, in two subjects

Mr Bowen made two mistakes on the same day. In **maths** he wrote

$$x + x = x^2$$

In **science** he said: "O₂ has two atoms, so it is a compound."

(a) Write the correct answer to $x + x$. _____

(b) Is O₂ an element or a compound? _____

(c) These two mistakes are the **same** mistake. Explain what it is, in full sentences.

E5 And he made it again

A week later Mr Bowen wrote these two lines, one in each book.

Maths: $4(x + 3) = 4x + 3$

Science: 6CO₂ is 6 carbon atoms and 2 oxygen atoms.

(a) Write the maths line correctly. _____

(b) Write the science line correctly. _____

(c) What did he forget to do, both times? Answer in one sentence.
